

Hardware Assessment Report

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Scan Date: April 26, 2026 at 04:17 PM



Estimated Lifespan: 1-2 years

System Information

Hostname: Sajan

OS: Microsoft Windows 11 Home Single Language 11

CPU: Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz

RAM: 15.83 GB

Overall System Health: 55/100 (Grade C)

Estimated Remaining Lifespan: 1-2 years

Critical Issues Detected:

- GPU: D (37/100) — Replace soon, degraded performance

Component Health Summary

Component	Score	Grade	Status
Intel(R) Core(TM) i5-10210U CPU @ 1	77	B	No action needed
15.83 GB (2 modules)	70	B	No action needed
INTEL SSDPEKNW512G8H	68	B	No action needed
Intel(R) UHD Graphics	37	D	Replace soon, degraded performance
HP 8656	67	B	No action needed
Internal Battery	84	A	No action needed

CPU — Intel(R) Core(TM) i5-10210U CPU @ 1.60GH

Grade B (77/100)

Part/Serial: BFEBFBFF000806ECStatus: ■ OK

→ Reapply thermal paste to reduce operating temperatures

→ Clean CPU heatsink and fan to prevent thermal throttling

RAM — 15.83 GB (2 modules)

Grade B (70/100)

Part/Serial: M471A1K43BB0-CPBStatus: ■ OK

→ Run MemTest86 to verify memory stability and rule out bad sectors

→ Ensure RAM is configured in dual-channel mode for optimal bandwidth

Storage — INTEL SSDPEKNW512G8H

Grade B (68/100)

Part/Serial: 0000_0000_0100_0000_E4D2_5CA4_E3D2_...Status: ■ OK

→ Defragment the drive regularly to maintain read/write performance

→ Monitor S.M.A.R.T. values weekly to detect early signs of failure

GPU — Intel(R) UHD Graphics

Grade D (37/100)

Part/Serial: Intel(R) UHD Graphics FamilyStatus: ■ REPLACE_SOON

→ Update GPU drivers to the latest stable version

→ Clean GPU fans and heatsink to prevent thermal throttling

Motherboard — HP 8656

Grade B (67/100)

Part/Serial: PJGGP00WBE8117Status: ■ OK

→ Update BIOS firmware to the latest version for compatibility fixes

→ Visually inspect capacitors for bulging or leaking (sign of failure)

Battery — Internal Battery

Grade A (84/100)

Part/Serial: N/ASTatus: ■ OK

→ Calibrate the battery every 3 months by fully charging and discharging

→ Keep charge levels between 20-80% for daily use to extend cycle life

Overall Benchmark Score: 92/100

CPU Compute

100/100



Baseline: 5000 ops (Intel i5-12400 class)

Measured: 135644 operations in 5s

Disk I/O

100/100



Baseline: 500 MB/s (NVMe SSD class)

Measured: 499.91 MB/s

Memory Bandwidth

67/100



Baseline: 10000 MB/s (DDR4-3200 class)

Measured: 6678.45 MB/s

Category	Status	Bottlenecks
Web Development	Fully Supported	None
Desktop Development	Minimum Requirements Only	Recommended 16GB RAM (has 15.83GB); Recommended 6 CPU cor...
Mobile Development	Minimum Requirements Only	Recommended 16GB RAM (has 15.83GB); Recommended 6 CPU cor...
Databases	Minimum Requirements Only	Recommended 16GB RAM (has 15.83GB)
Virtualization	Minimum Requirements Only	Recommended 32GB RAM (has 15.83GB); Recommended 8 CPU cor...
AI & Data Science	Not Supported	Recommended 32GB RAM (has 15.83GB); Recommended 8 CPU cor...
CAD & Engineering	Not Supported	Recommended 32GB RAM (has 15.83GB); Recommended 8 CPU cor...
Game Development	Not Supported	Recommended 32GB RAM (has 15.83GB); Recommended 8 CPU cor...
Video Editing	Not Supported	Recommended 32GB RAM (has 15.83GB); Recommended 8 CPU cor...
Basic Office & Browsing	Fully Supported	None

✓ No Hardware Tampering Detected

All hardware components match the registered baseline.

Total Recommendations: 13

- [MEDIUM] GPU
Replace soon, degraded performance
- [HIGH] System
Recommended 32GB RAM (has 15.83GB)
- [HIGH] System
Recommended 8 CPU cores (has 4)
- [HIGH] System
Needs 4096MB VRAM (has 1024.0MB)
- [HIGH] System
Needs 2048MB VRAM (has 1024.0MB)

Total Estimated Upgrade Cost

Rs. 26,200 INR (~\$295 USD)

Prices as of 2026-04-07 | Sources: Amazon.in, MDComputers, PrimeABGB

Priority	Component	Current	Recommended Upgrade	■ INR	\$ USD
OK	RAM	15.83GB DDR4	DDR4 16GB 3200MHz	Rs. 4,200	\$50
✓ GOOD	Storage	476.94GB HDD (INTEL SS	512GB SATA SSD	Rs. 4,000	\$45
OK	CPU	Intel(R) Core(TM) i5-1	Intel i5 (12th/13th Gen)	Rs. 18,000	\$200
			TOTAL	Rs. 26,200	\$295

Upgrade Justification

- RAM: Upgrade from 15.83GB to 16GB DDR4
Source: Amazon.in (<https://www.amazon.in/s?k=ddr4+16gb+ram>)
- Storage: Replace 476.94GB HDD with 512GB SATA SSD (3-5x speed boost)
Source: MDComputers (<https://mdcomputers.in/ssd?capacity=512gb>)
- CPU: Upgrade from 4-core/2112.0MHz to modern mid-range CPU
Source: MDComputers (<https://mdcomputers.in/processors?brand=intel&tier=i5>)

* Prices are approximate and may vary. Consult your vendor for exact quotes. Currency rate: 1 USD = ■92.0